

Alidade wire harnessing Enclosure								
Reference ELMCOoverview.vsd								
7-Apr-06								
AWE1 Input power wires through pedestal Belden 8527/8520								
	A	Description	B	Comment	Color	Size	Twist	Length
Wires	1	120VAC			white	#12		
	2	Return			black	#12		
	3	Ground			green	#14		
Ends	A	bare wire		screw terminal block behind pie-plate				
	B	39860-0103		Molex terminal on Alidade Relay Board				
AWE2 Transformer								
	A	Description		Comment	Color	Size	Twist	Length
Wires	1	120VAC		Wires on toroid	white	#14		
	2	Return			black	#14		
	3							
	4	100VAC			red	#14		
	5	Return			red	#14		
End	A	39860-0105		Molex terminal on alidade relay card				
AWE3 Power to motors Volex 17539								
	A	Description	B	Comment	Color	Size	Twist	Length
Wires	1	110VAC			black	#16		21" w/ 5" stripback
	2	Return			white	#16		
	3	Ground			green	#16		
Ends	A	39860-0104		Molex terminal on alidade relay board				
	B	IEC-320		plug on drive box		7.5" rubber air hose		
AWE4 Power for control box Volex 17539								
	A	Description	B	Comment	Color	Size	Twist	Length
Wires	1	120VAC			white	#16		19.5" w/ 5" stripback
	2	Return			black	#16		
	3	Ground			green	#16		
Ends	A	39860-0103		Molex terminal on alidade relay board				
	B	IEC-320		plug on drive box		18" rubber air hose		
AWE5 Power for rim power box (also called AR1) wire								
	A	Description	B	Comment	Color	Size	Twist	Length
Wires	1	120VAC			black	#14		
	2	Return			white	#14		
	3	Ground			green	#14		
Ends	A	39860-0103		Molex terminal on alidade relay board				
	B	ParlBlade U Gnd		connector in rim power box				
AWE6 15-wire cable from alidade relay board to drive box CablesToGo custom								
	A	Description	B	Comment		Size		Length
Wires/pins	1	az brakes +24V	1	1&9 are controlled with the main drives transformer power				
	2	Ground	2					
	3	24V aux	3			all #24		23"
	4	Ground	4					
	5	drive enable +	5	5&13 are connected through relay contacts which are closed when the main drives transformer power relay is latched				
	6	Ground	6					
	7	Az cw limit	7	7&15 are connected to 8 via NC switches				
	8	Az limit common	8					
	9	el brakes +24V	9					
	10	Ground	10	Belden 9947 maybe				
	11	24V aux	11	awaiting quote from cablestogo.com				
	12	Ground	12					
	13	drive enable -	13					
	14	Ground	14					
	15	Az ccw limit	15					
Ends	A	D15P		Dsub to alidade relay board				
	B	D15S		Dsub to drive box	connects to AWD2A			

AWE7									25-wire cable from alidade relay board to control box									CablesToGo custom								
			A	Description	B			Comment									Length									
Wires/pins				1	brakes +24V	1			1,3,5,11,14,16,18, 24 are power and control signals from the																	
				2	Ground	2			control box																	
				3	24V aux	3																				
				4	Ground	4																				
				5	5V sensor +	5																				
				6	Ground	6																				
				7	float in	7			7,8,9,10,12,13,20,21,22,23,25 are logic and sensor signals																	
				8	T input1 -	8			returning to the control box																	
				9	T input2 -	9																				
				10	Az brake sense	10																				
				11	Latch	11			main drives transformer power is latched by holding 24 logic high																	
				12	Pot 1	12			while pulsing 11 logic low-high-low																	
				13	Pot wiper	13			wrap pot is wired to relay board such that resistance between 13																	
				14	brakes +24V	14			and 25 increases when azimuth rotates cw																	
				15	Ground	15																				
				16	24V aux	16																				
				17	Ground	17																				
				18	5V sensor -	18																				
				19	Ground	19																				
				20	T input1 +	20																				
				21	T input2 +	21																				
				22	electronics reset	22																				
				23	El brake sense	23																				
				24	Unlatch	24																				
				25	Pot 2	25																				
Ends				A	D25S	Dsub to alidade relay board																				
				B	D25P	Dsub to control box																				
AWE8									motor power and brakes for azimuth									part of motor								
			A	Description	B, C			Comment			Color			Size			Twist			Length						
Wires/pins				1	W1	BW						purple			#14											
				2	W2																					
				14	W3																					
				15	W4																					
				3	N/C																					
				16	N/C																					
				4	V1	BV						brown			#14											
				5	V2																					
				17	V3																					
				18	V4																					
				6	N/C																					
				19	N/C																					
				7	U1	BU						blue			#14											
				8	U1																					
				20	U1																					
				21	U1																					
				9	N/C																					
				22	N/C																					
				10	G1	BGND						green			#14											
				11	G2																					
				23	G3																					
				24	G4																					
				12	BP	C						blue			#16											
				13	BM	C						blue			#16											
				25	SHIELD	N/C						shield			shield											
Ends				A	D25P	at driver board																				
				B	at motor																					
				C	39-01-3023/39-00-0041	wires part of brake, but with molex																				

		mate: 39-01-2020/39-00-0039						
AWE9	encoders and hall's for azimuth				part of motor			
	A	Description	B	Comment	Color	Size	Twist	Length
Wires/pins	1	N/C		GND on board	N/C			
	2	U			brown			
	3	V			gray			
	4	W			white			
	5	GND			black		V	
	6	X			purple		Z	
	7	B			green		Y	
	8	A			blue		X	
	9	thermal sensor		GPIN5	orange		W	
	10	GND thermal sensor		GND	orang/white		W	
	11	Vcc (+5V)			red		V	
	12	N/C		GND on board	N/C			
	13	/X			purple/blk		Z	
	14	/B			green/blk		Y	
	15	/A			blue/blk		X	
Ends	A	D15P	at driver board		incline 45 towards pin 15			
	B	M						
AWE10	extender for motor power and brakes for elevation				Belden 83352 + #18 hookup wire			
	A	Description	B	Comment	Color	Size	Twist	Length
Wires/pins	1	W1	4		Black	#16		158
	2	W2						
	14	W3						
	15	W4						
	3	N/C	3					
	16	N/C						
	4	V1						
	5	V2						
	17	V3						
	18	V4						
	6	N/C	2					
	19	N/C						
	7	U1						
	8	U2						
	20	U3	5					
	21	U4						
	9	N/C						
	22	N/C						
	10	G1	8		White	#16		
	11	G2						
	23	G3						
	24	G4						
	12	BP	9		Blue hook-up	#18		
	13	BM						
	25	SHIELD						
Ends	A	D25P	plugs into D25S on board					
	B	bare wire	MBTO terminal strip					

AWE11 extender for encoders and hall's for elevation Belden 8307								
	A	Description	B	Comment	Color	Size	Twist	Length
Wires/pins	1	N/C		GND on board				158
	2	U	2		Brown		T	
	3	V	3	Note: take care to	Black		T	
	4	W	4	get the right black	Blue			
	5	GND	5	for twisted pair	Black		V	
	6	X	14	as indicated	Orange		Z	
	7	B	12		White		Y	
	8	A	10		Yellow		X	
	9	thermal sensor	7	GPIN5	Green		W	
	10	GND thermal sensor	8	GND	Black		W	
	11	Vcc (+5V)	6		Red		V	
	12	N/C		GND on board				
	13	/X	15		Black		Z	
	14	/B	13		Black		Y	
	15	/A	11		Black		X	
Ends	A	D15P	Dsub pin on end of elevation sensor extender			incline 45 towards pin 15		
	B	bare wire	HTE terminal strip					
AWE12 encoder for azimuth (P/N: R176H-03600D-5L10-AT40SP-24MN)								
	A	Description	B	Comment	Color	Size	Twist	Length
Wires/pins	1	A			yellow			30
	2	B			green			
	3	IND			blue			
	4	Shield			shield			
	5	+V			red			
	6	/A			brown			
	7	/B			orange			
	8	/IND			white			
	9	COMMON			black			
Ends	A	D9P	Dsub pin on end of encoder cable			incline 45 towards pin 1		
	B	--	cable part of encoder					
AWE13 encoder for elevation (P/N: 9220S03600D5L10D99SP04EA)								
	A	Description	B	Comment	Color	Size	Twist	Length
Wires/pins	1	A			yellow			
	2	B			green			
	3	IND			blue			
	4	Shield			shield			
	5	+5V			red			
	6	/A			brown			
	7	/B			orange			
	8	/IND			white			
	9	COMMON			black			
Ends	A	D9P	Dsub pin on end of encoder cable					
	B	--	cable part of encoder					
AWE14 battery for control box wire								
	A	Description	B	Comment	Color	Size	Twist	Length
Wires/pins	1	+12V	1	use 1-#18	red	18		22
	2	+12V	1	or 3-#22	red	same wire		22
	3	N/C						
	4	GND	2	use 1-#18	black	18		19
	5	GND	2	or 3-#22	black	same wire		19
	6	+12V	1		red	same wire		22
	7	N/C						
	8	N/C						
	9	GND	2		black	same wire		19
Ends	A	D9S	at controller box					
	B	Spade	at battery					

AWE15	Signal from relay board to pie-plate			LappUSA (Olflex) 900P Cable, 3 conductor #24, 811442				
	A	Description	B	Comment	Color	Size	Twist	
Wires	1	M-stop	1		white	#24		
	2	reset	2		brown	#24		
	3	GND	3		green	#24		
Ends	A	Molex 43645-0300/43030-0009	on alidade relay board (Jameco 338360/337738)					
	B	bare	terminal behind pie plate					
AWE16	signal from az oil float			wire				
	A	Description	B	Comment	Color	Size	Twist	
Wire	1	Float 1			red	#24		
	2	Float 2			red			
Ends	A	Molex 43645-0200/43030-0009	on alidade relay board (Jameco 338351/337738)					
	B	--	w					
AWE17	Signal from temperature sensor to relay board			wire				
	A	Description	B, C	Comment	Color	Size	Twist	
Wires	1	+5V	B1		Red	#24		
	2	+5V	C1		Orange	#24		
	3	T (T1)	B3		Yellow	#24		
	4	T (T2)	C3		Yellow	#24		
	5	GND	B2		Black	#24		
	6	GND	C2		Black	#24		
Ends	A	Molex 43025-0600/43030-0009	to alidade relay board					
	B	solder to LM35	Lm-35 Device in TO-220					
	C	solder to LM35	Lm-35 Device in TO-220					
AWE18	Signal from az wrap potentiometer			wire				
	A	Description	B	Comment	Color	Size	Twist	
Wires	1	POT1	1	pin 12 (AWE7)	black	#24		
	2	POT WIPER	2	pin 13 (AWE7)	yellow			
	3	POT 2	3	pin 25 (AWE7)	green			
Ends	A	Molex 43645-0300/43030-0009	on alidade relay board					
	B	bare wire	posts on pot					
AWE19	Signal from elevation level sensor and thermal sensor			Belden 9611				
	A	Description	B	Comment	Color	Size	Twist	Length
Wires	1	+5V	10	assume A has odd/	brown	#24		
	2	+5V	11	even rows	red			160"
	3	T (T3)	12		white			
	4	Level (T4)	13	cable may change	green			
	5	GND	14	to Belden 9940	black			
	6	GND	15		blue			
Ends	A	Molex 43025-0600/43030-0009	on relay board					
	B	bare wire	MBTO terminal strip					